Section - 9

TECHNICAL SPECIFICATION FOR PANELS

SCOPE : Refer Clause-1 of Section-I, General

1.1 This Specification provides for design, engineering, manufacture, assembly, shop testing, inspection and testing before despatch, packing, forwarding and delivery at site CIP destination (in case of Indian Bidders) and CIF Indian Port at the discretion of the Purchaser (in case of Foreign Bidders) of Control, relay, alarm and annunciation panels for Transformer HV & LV, feeder, and Bus bar for substations of various voltages and ratings.

1.2 It is not the intent to specify completely herein all details of the design and construction of equipments. However, the equipment shall conform in all respects to high standards of engineering, design and workmanship and shall be capable of performing in continuous commercial operation acceptable to the purchaser, who will interpret in a manner the meanings of drawings and specification and shall have the power to reject any material which in his judgement is not in accordance therewith. The equipment offered shall be complete with all components necessary for its effective and trouble free operation. Such components shall be deemed to be within the scope of Bidder's supply, irrespective of whether these are specifically brought in this specification and/or the commercial order or not.

2.0 STANDARDS:

All equipment provided under this specification shall in general conform to the latest revision/amendment etc. of the following:

Sl. No.	Standard	Title
1.	IS 3231 IEC 255 Part I to III BS 142	Electrical relays for power system protection
2.	IS 1248 IS 2419 BS 89	Indicating instruments
3.	IS-13010,IS 722(Part Ito IX) IEC 521 BS 3T	Energy meters
4.	IS 4237	General requirement of switch-gear and control gear for voltages not exceeding 1 **kV**
5.	IS 375	Marking arrangement for switch-gear, bus bars, main connection and auxiliary wiring

S.No.	Standard	Title
6.	IS 8686 IEC-255Part V&VI IEC 801-4	Specification for static protective relays and tests.
7.	IEC 337 IEC 337-1 IS 6875	Control switches (LV switching devices for control and auxiliary circuits
8.	IS 2516	MCCB sand load break switches having a break capacity of 30 **kV**A rms at 415 volts
9.	IS 4047	Fuse switching units
10.	IS 2208	HRC fuse links
11.	IS 2705 IEC-185	Current transformers
12.	IS 3156 IEC-186	Potential transformers
13.	IS 8623	Factory built assemblies of switch gear and control gear for voltages upto and including 1000V AC and 1200V DC.
14.	IEC 439	Low voltage switch gear and control gear assemblies
15.	IS 4237	General requirement of switch- gear and control gear for voltages not exceeding 1000V
16.	IS 2147	Degree of protection provided by enclosures for low voltage switch gear and control gear
17.	IS 5082	Aluminum alloy bars
18.	IS 1897	Copper strips for electrical purpose
19.	IS 3618 IS 6005	Code of practice for phosphate treatment of iron and steel
20.	IS 4064	Air break switches, air break dis-connectors, air break disconnectors and fuse combination units for voltages not exceeding 1000V AC or 1200V DC

Sl. No.	Standard	Title
21.	IS 9224 Part II	HRC catridge fuses
22.	IS 5578	Guide for marking of insulated conductors.
23.	IS 2032	Graphical symbols Ref- abbreviations
		Name and Address
IEC		Institution Electro Technical Bureau. Central de la commission Electro Technique International. 1. Rue de verembe Geneva Switzerland.
ISO		International Organization for Standardization. Danish board of standardization. Danish standardizing sruat Aurehoeqvaj - 12 DK - 2900, Heel prup DENMARK
IS		Indian Standard Bureau of India. Standards Manak Bhavan, 9, Bahadur Shah Zafar Marg. New Delhi - 110 002. INDIA

Equipment meeting with the requirements of other authoritative Standards, which ensure equal or better performance than the standards mentioned above, shall also be considered. When the equipment offered by the Bidder conforms to other standards, salient points of difference between standards adopted and the standards specified in this specification shall be clearly brought out in the relevant schedule. Four copies of such standards with authentic translation in English shall be furnished along with the offer.

5.0 GENERAL TECHNICAL REQUIREMENTS:

5.1.0 panels: Detailed description:

The control and relay panels shall be of Simplex type. Each Simplex panel shall consist of a vertical front panel with equipment mounted thereon and having wiring access from the rear. It shall have double leaf door with lift off hinges at the back for panels of width more than 800 mm. Doors shall have handles with built-in locking facility.

5.1.1 Constructional Features:

a) Control and relay panels shall be of Simplex type. It is the responsibility of the contractor to ensure that the equipment specified and such unspecified complementary equipment required for completeness of the protective/control schemes be properly accommodated in the panels without congestion and if necessary provide panels with larger dimensions. No price increase at a later date on this account shall be allowed.

b) Panels shall be completely metal enclosed and shall be dust, moisture and vermin proof. The enclosure shall provide a degree of protection not less than IP-44 in accordance with IS:2147.

c) Panels shall be free standing, floor mounting type and shall comprise structural frames enclosed completely with specially selected smooth finished, cold rolled sheet steel of thickness not less than 3 mm for weight bearing members of the panels such as bass frame, front sheet and door frames, and 2.0 mm for sides, door top and bottom portions. There shall be sufficient reinforcement to provide level surfaces, resistance to vibration and rigidity during transportation and installation.

d) All doors, removable covers and panels shall be gasketed all around with neoprene gaskets. Ventilating louvers, if provided shall have screens and filters. The screens shall be made of either brass or GI wire mesh.

e) Design, materials selection and workmanship shall be such as to result in neat appearance inside and outside with no welds, rivets or bolt head apparent from outside, with all exterior surfaces true and smooth.

f) All necessary anchor bolts and other materials required for foundation of panels shall be supplied.

g) Cable entries to the panel shall be from the bottom. The plates of the panel shall be fitted with removable gland plates for fixing the cable glands. Necessary number of cable glands of sizes to suit owner's external cables to the panels shall be supplied by the bidder. Cable glands shall be screwed type, and shall be suitable for PVC cables. Cable gland plate fitted in the bottom of the Panel shall be connected to the earthing Panel/Station through a flexible braided copper conductor rigidly.

h) Relay panels of modern, modular construction would also be acceptable.

i) The total height of the panel shall be 2250mm. The depth of the panel shall be 800mm. The panels shall have suitable width required for mounting of all the equipments on the front panel.

j) The panels shall match with the existing panels in APTRANSCO's substations, the details of which will be furnished to the successful bidder. Matching includes height, depth of the panels, Colour matching and mimic matching and arrangement of equipment on the front.

k) The width and depth of panel shall be well designed for giving sufficient room and space for ease of working and safety.

5.1.2 Mounting

a) All equipment on and in panels shall be mounted and completely wired to the terminal blocks ready for external connection. The equipment on front of panel shall be mounted flush.

b) Equipment shall be mounted such that removal and replacement can be accomplished individually without interruption of service to adjacent devices and are readily accessible without use of special tools. Terminal marking shall be clearly visible.

c) The contractor shall carry out cutouts, mounting and wiring of the free issue items supplied by others, if any which are to be mounted in this panel in accordance with the corresponding equipment manufacturer's drawings. Cutouts if any, provided for future mounting of equipment shall be properly blanked off.

d) The center lines of switches, push buttons and indicating lamps shall be not less than 750mm from the bottom of the panel. The center lines of relays, meters and recorders shall be not less than 450mm from the bottom of the panel.

e) The center lines of switches, push buttons and indicating lamps shall be matched to give a neat and uniform appearance. Likewise the top lines of all meters, relays and recorders etc., shall be matched.

f) No equipment shall be mounted on the doors.

g) All the equipment connections and cabling shall be designed and arranged to minimize the risk of fire and damage which may be caused by fire.

5.1.3 Panel Internal Wiring and other Accessory Equipment :

a) Panels shall be supplied completely with interconnecting wiring provided between all electrical devices mounted and wired in the panels and between the devices and terminal blocks for the devices to be connected to equipment outside the panels. When panels are arranged to be located adjacent to each other all inter panel wiring and connections between the panels shall be furnished and the wiring shall be carried out internally. These adjacent inter panel wiring shall be clearly indicated in the drawing furnished by the contractor.

b) All wiring shall be carried out with 1100 V grade, single core, stranded copper conductor wires with PVC insulation and shall be heat resistant grade and vermin and rodent proof. The minimum size of the stranded copper conductor used for internal wiring shall be as follows:

i) All circuits except current transformer circuits one multi strand 1.5 sq.mm per lead
ii) Current transformer circuit: one 2.5 sq.mm per lead. The minimum number of strands per conductor shall be three.

c) All internal wiring shall be securely supported, neatly arranged, readily accessible and connected to equipment terminals and terminal blocks. Wiring gutters and troughs shall be used for this purpose.

d) Auxiliary bus wiring for AC and DC supplies, voltage transformer circuits, annunciation circuits and other common services shall be provided near the top of the panels running throughout the entire length of the panels.

e) Wire termination shall be made with solder less crimping type and tinned copper lugs which firmly grip the conductor and insulation. Insulated sleeves shall be provided at all the wire terminations. Engraved core identification plastic ferrules marked to correspond with panel wiring diagram shall be fitted at both ends of each wire. Ferrules shall fit tightly on the wire and shall not fall off when the wire is disconnected from terminal blocks.

f) All wires directly connected to trip circuit breaker or device shall be distinguished by the addition of red coloured unlettered ferrule. Numbers 6 and 9 shall not be included for ferrule purposes.

g) Longitudinal troughs extending throughout the full length of the panel shall be preferred for inter panel wiring. Interconnections to adjacent panel shall be brought out to a separate set of terminal blocks located near the slots or holes meant for taking the inter connecting wires. Arrangements shall permit easy inter connections to adjacent panels at site and wires for this purposes shall be provided by Contractor looped and bunched properly inside the panel.

h) Contractor shall be solely responsible for the completeness and correctness of the internal wiring and for the proper functioning of the connected equipment

i) For Inter panel wiring , terminations shall be done on bottom side of panel.

j) The indicating lamp shall be of LED type with sufficient brightness. Coloured LED lamps shall be used for appropriate indication.

5.1.4 Terminal Blocks

a) All internal wiring to be connected to the external equipment shall terminate on terminal blocks preferably vertically mounted on the side of each panel. Terminal blocks shall be

1100V grade and have 10 amps continuous rating moulded piece complete with insulated barriers, stud type terminals, washers, nuts and lock nuts. Terminal block design shall include a white fibre markings strip with clear plastic slip-on/ clip-on terminal covers. Markings on the terminal strips shall correspond to wire number and terminal numbers on the wiring diagrams.

b) Terminal blocks for current transformer and voltage transformer secondary leads shall be provided with test links and isolating facilities. Also current transformer secondary leads shall be provided with short circuiting and earthing facilities.

c) At least 20% spare terminals shall be provided on each panel and these spare terminals shall be uniformly distributed on all terminal blocks.

d) Unless otherwise specified, terminal blocks shall be suitable for connecting the following conductors of purchasers cable on each side:

i) All circuits except current transformer circuits: minimum of one of 2.5 sq.mm copper

ii) All CT circuits: minimum of two of 2.5 mm square copper and all PT circuits minimum of four of 2.5 sq.mm copper

e) There shall be a minimum clearance of 250 mm between the first row of terminal blocks and associated cable gland plate. Also the clearance between two rows of terminal blocks shall be minimum, of 150 mm.

f) Arrangement of the terminal block assemblies and the wiring channel within the enclosure shall be such that a row of terminal blocks is run parallel and in close proximity along each side of wiring duct to provide for convenient attachment of internal panel wiring. The side of the terminal block opposite the wiring duct shall be reserved for purchaser's external cable connection. An adjacent terminal block shall also share this field wiring corridor. A steel strip shall be connected between adjacent terminal block rows at 450 mm intervals for support of incoming cables.

g) The number and sizes of the Purchaser's multi-core incoming cable will be furnished to the Contractor after placement of the order. All necessary cable terminating accessories such as gland plates, packing glandser, crimp type tinned copper lugs supporting clamps and brackets, wiring troughs and gutters etc., for purchaser's cable shall be included in contractor's scope of supply.

5.1.5 Painting

a) All sheet steel work shall be phosphated in accordance with the IS:6005 "Code of practice for phosphating iron and steel"

b) Oil, grease, dirt and sweat shall be thoroughly removed by emulsion cleaning.

c) Rust and scale shall be removed by picking with dilute acid followed by washing with running water rinsing with a slightly alkaline hot water and drying.

d) After phosphating thorough rinsing shall be carried out with clean water followed by final rinsing with dilute dichromate solution and oven drying.

e) The phosphate coating shall be sealed with application of two coats of ready mixed, stoved type zinc chromate primer. The first coat may be "flash dried" while the second coat shall be stoved.

f) After application of the primer, two coats of finishing synthetic enamel paint shall be applied, each coat followed by stoving. The second finishing coat shall be applied after completion of tests. The exterior Colour of the paint shall be selected by the purchaser at a later date.

g) Each coat of primer and finished paint shall be of a slightly different shade to enable inspection of the painting.

h) A small quantity of finished paint shall be supplied for minor touching up required at site after installation of the panels.

i) In case the bidder proposes to follow any other established painting procedure like electrostatic painting, the procedure shall be submitted along with offer/bid for purchaser's review and approval.

5.1.6 Mimic Diagram

a) Coloured mimic diagram and symbols showing the exact representation of the system shall be provided in the front of control panels.

b) Mimic diagram shall be made preferably of anodized aluminum or plastic of approved fast Colour material which shall be screwed on to the panel and can be easily cleaned. Painted overlaid mimic is also acceptable. The mimic bus shall be 2 mm thick. The width of the mimic bus shall be 10 mm for bus bars and 7 mm for other connections.

c) Mimic bus Colour will be decided by the Purchaser and shall be obtained from the Purchaser by the successful Bidder.

d) When semaphore indicators are used for isolator position they shall be so mounted in the mimic that the isolator closed position shall complete the continuity of mimic.

e) Indicating lamp, one for each phase for each bus shall be provided on the mimic to indicate bus charged condition.

5.1.7 Name plates and markings:

a) All equipment mounted on front and rear side as well as equipment mounted inside the panels shall be provided with individual name plates with equipment designation engraved. Also on the top of each panel, on front as well as rear side, large and bold name plates shall be provided for circuit/feeder designation.

b) All front mounted equipment shall be also provided at the rear with individual name plates engraved with tap numbers corresponding to the one shown in the panel internal wiring to facilitate easy tracing of the wiring. The name plates shall be mounted directly by the side of the respective equipment in a visible manner and shall not be hidden by the equipment wiring.

c) Name plates shall be made of non-rusting metal or 3 ply lami-coid. Name plates shall be black with white engraving lettering. The name plates inscription and size of name plates and letters shall be submitted to the Purchaser for approval.

d) Each instrument and meter shall be prominently marked with the quantity measured e.g. kV, A, MW, etc. All relays and other devices shall be clearly marked with manufacturer's name, manufacturer's type, serial number and electrical rating data.

e) Each switch shall bear clear inscription, identifying its function eg. "BREAKER" "52 A" "AMMETER" "SYNCHRONISING" etc., similar inscriptions shall also be provided on each device whose function is not otherwise identified. If any switch device does not bear this inscription separate name plate giving its function shall be provided for it. Switch shall also have clear inscription for each position indicating eg. "TRIP-NEUTRAL-CLOSE", "ON-OFF", "R-Y-B-N-OFF" etc.,

5.1.8 Miscellaneous Accessories and Supporting Steel:

a) Plug point-A AC 240 Volts, single phase, 50 HZ AC plug point shall be provided in the interior of each cubicle with "ON-OFF" Switch for connection of hand lamps.

b) Interior lighting: Each panel shall be provided with a fluorescent lighting fixture rated for 240 volts, single phase, 50 HZ supply for the interior illumination of the panel during maintenance. The fittings shall be complete with switch fuse unit and switching of the lighting shall be controlled by the respective panel door switch.

c) Each control panel shall be provided with necessary arrangements for receiving, distributing, isolating and fusing of DC and AC supplies for various control signaling, lighting and space heater circuits. MCBs for the incoming circuits and fuses for sub-circuits shall be provided. Selection of the main and sub-circuit fuse rating shall be such as to ensure selective clearance of sub-circuit faults. Potential circuit for relaying and metering shall be protected by fuses. All fuses shall be HRC cartridge type conforming to IS:2208 mounted on plug-in type fuse bases. All accessible live connection to fuse bases shall be adequately shrouded. Fuses shall have operation indicators for indicating blown fuse condition. Fuse carrier base shall have imprints of the fuse rating and voltage.

d) Supporting steel:

All necessary foundation anchor bolts and other parts of supporting and fastenings, of the panels shall be supplied by the contractor.

5.1.9 Earthing:

a) All panels shall be equipped with an earth bus securely screwed. Location of earth bus shall ensure no radiation interference for earth systems under various switching conditions of isolators and breakers. The material and the sizes of the bus bar shall be atleast 25 x 6 mm copper flat.
When several panels are mounted adjoining each other, the earth bus shall be made continuous and necessary connectors and clamps for this purpose shall be included in the scope of supply of contractor. Provision shall be made for extending the earth bus bars to future adjoining panels on either side.

b) Provision shall be made on each earth bus bars of the end panels for connecting Purchaser's earthing grid. Necessary terminal clamps and connectors for this purpose shall be included in the scope of supply of contractor. The wire or screens should be clearly bonded and earthed at the gland plate.

c) All metallic cases of relays, instruments and other panel mounted equipment shall be connected to the earth bus preferably by independent copper wires of size not less than 2.5 sq.mm. The colour code of earthing wires shall be green. Earthing wire shall be connected on terminals with suitable clamp connectors and soldering shall not be permitted.

d) Looping of earth connections which would result in loss of earth connection to other devices where the loop is broken shall not be permitted. However, looping of earth connections between equipment to provide alternative paths to earth bus shall be provided.

e) VT and CT secondary neutral or common lead shall be earthed at one place only at the terminal blocks where they enter the panel. Such earthing shall be made through links so that earthing may be removed from one group without disturbing continuity of earthing system for other groups.

5.8 LTAC PANELS: DETAILED DESCRIPTION

5.8.1 The LTAC panel shall comprise of sheet steel clad dust and vermin proof compartmentalized cubicle type panel. The panels shall be of floor mounting and free standing type and suitable for indoor installation.

The panels shall be dust and vermin proof. Industrial LT AC panels are not acceptable, panels shall consist of cubicles bolted together to form a flush front switch board with door on rear, compartmental doors on front with cable entry from bottom and with the following specifications :

Sheet steel thickness - 2 mm CRCA sheet steel

Height	-	2000 mm
Depth	-	600 mm
Width	-	As required to accommodate incoming, out going and spare circuits
Base channel	-	75 x 40 mm fabricated out of 3 mm sheet steel
Door interlock	-	Each compartment which incorporates switch fuse units or load break switches or MCCB is to be with door interlock.
Barrier between feeders	-	Each compartment is to be segregated and closed on all sides except for the bus bar/cable connector opening.
Switch connections	-	Feeders above 63 A to be connected by bus bar links. Switches of 63 A and below to be connected by PVC copper wires using suitable crimped, copper lugs of Dowells make.
Bus bars	-	Three Phase Neutral aluminum PVC sheathed Phase size 62 x 10 mm Neutral 31 x 10 mm Aluminium alloy used for bus bars shall be equivalent to E91E quality of IS 5082 Bus bars to be supported on unbreakable non-hygroscopic insulated supports. Bus bars to be fully shrouded.
Terminals for outgoing	-	Suitable bus bar links for connections 100 A and higher size feeders and terminals for 63 A and smaller size feeder to be provided.
Incoming MCCB	-	MCCB with clearly marked ON-OFF positions with rotary mechanism provided with front handle shaft and door interlock.
Switch fuse units	-	Combination switch fuse units double break mechanism AC 23 duty rating with door inter- lock and pad locking mechanism.

5.8.2 Panels shall be completely factory wired, ready for connecting to the equipment . Control wiring shall be done with 2.5 Sq.mm 1100V grade PVC insulated stranded copper cables. Panels shall be given two coats of light gray paint finish after thorough pre-treatment and application of corrosion resistant primer.

Earthing shall be done for all the equipment and the panel. The minimum size of earthing bus shall be 3x25 mm copper.

Construction of panels shall be such that it is possible to add more incoming/outgoing circuits by suitable and easy extensions. The incoming and outgoing circuits shall be provided with detailed equipment schedule as given in Annexure

6.0 I. TESTS:

The bidder shall carryout routine tests on all associated equipment as per relevant Indian Standards/IEC Standards. The bidder shall submit along with the bid, one set of type test report for protective relays, instruments and panels, switches and other devices of the type and class being offered to asses the suitability of the equipment. The type tests must have been conducted not earlier than five years form the date of bid opening.

Control and relay panels shall be subjected to the following tests:

a) Mechanical operation tests

b) Verification of degree of protection as per IS:2147

c) High voltage test (2 kV for 1 minute) or as per relevant IS.

d) Insulation resistance test

e) Electrical control, interlock and sequential operation test

f) Verification of wiring and general arrangement as per approved schematic/G.A. diagrams.

II. SUPERVISION OF COMMISSIONING CHARGES :

The tenderer shall quote unit rates and lumpsum amount for supervision of commissioning charges for each item separately as given below :

No. of Days : 3 per panel / SS.
Manday rate / day

To and fro charges
Over/Time charges per hour

7.0 INSPECTION:

7.1 i) All routine tests shall be conducted at the time of inspection and inspection shall be made at the place of manufacturers unless otherwise specifically agreed upon by the manufacturer and purchaser at the time of purchase. The manufacturer shall offer the inspector representing the purchaser all reasonable facilities without charges to satisfy him that the panels/materials are manufacturers/supplied as per the specification/final approved drawings.

ii) The supplier shall give 15 days (for local supplier) 60 days(in case of foreign supplier) advance intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests.

iii) The panels shall be manufactured and offered for inspection as per the final approved General Arrangement Drawing and schematic drawing which will be binding both on supplier and purchaser.

iv) The panels shall not be despatched from its point of manufacture unless the same are satisfactorily inspected and tested.

iv) The acceptance of any material shall in no way release the supplier of his responsibility for meeting all the requirements of the specification and shall not prevent subsequent rejection of such material later found to be defective.

8.0 QUALITY ASSURANCE PLAN: Refer Clause-3 of Section-I, General

9.0 GUARANTEED TECHNICAL PARTICULARS:

The guaranteed technical particulars of the equipment offered shall be furnished by the bidder as per Volume-2B enclosed.

10.0 DOCUMENTATION:

10.1 The bidder shall furnish the dimensioned drawings along with the bid indicating the details like components, make, quantity, ranges of meters, relays etc., in case the bidder quotes different makes of relays shall furnish copies of wiring scheme and literature of all relays and sets of drawings and manuals shall be supplied along with each panel.

11.0 PACKING AND FORWARDING: Refer Clause-4 of Section-I, General

11.1 Name plate (giving details of equipment) in all equipments (i.e. relays, breaker switches semaphore indicating lamps, push buttons etc.) mounted on the panel shall be provided. Name plates of suitable size on metal/PVC acrylic/thick plastic sheets shall only be provided. Stickers in no case will be accepted. Large and bold name plates shall be provided for circuit/feeder designation. Name plates for AC fuses, DC fuses, resisters and other equipment mounted inside the panel shall be provided clearly indicating the purpose for which they are meant for.

Name plates of suitable size shall be provided on the top of the panel clearly indicating manufacturers name, purchase order number and date, year of manufacturer, serial number of the panel and type of panels. The panels shall be packed in suitable sized strong and weather resistant wooden cases / crates.

11.2 Suitable cushioning, protective padding or spacer shall be provided to prevent damage to or deformation of the components during transit and handling.

11.3 All packing cases shall be marked legibly and correctly with the suppliers name and consignee's address so as to ensure their safe arrival at their destination and to avoid possibility of goods being lost or wrongly dispatched on account of faulty or illegible marking.

12.0 COORDINATION:

12.1 Outdoor switch gear like isolators, breakers, CTs, PTs are being ordered separately and their characteristics will be furnished to the successful bidder, so that balanced systems of schemes of protection can be coordinated; any matching current transformers required may be provided and the scheme of protection made reliable in operation. The current transformers are of 1A/0.5 Amps. Secondary and potential transformers' secondary voltages are: 110 V / √3 and 110 V / √3 & 110 V

12.2 The bidder shall also prepare and supply the coordination wiring diagrams between the control and relay panels and control switch gear as part of this contract. The control and wiring diagrams of the switch gear will be supplied to the successful bidder.

12.3 The bidder shall coordinate in all respects with the manufacturers of associated equipment and shall freely exchange all relevant information to ensure satisfactory and economic over all design. No extra charge shall be payable for these services.

12.4 Erection of equipment called for in this specification will be undertaken by the purchaser. The bidder shall furnish full detailed instructions, complete with drawings and leaflets to enable satisfactory erection and commissioning of the plant by the departmental staff. Such instructions should reach the purchaser sufficiently in advance of the receipt of the equipment in order that all the preliminary arrangement for the erection of the equipment can be made by the department in advance so that there will be no delay in regard to erection as soon as the equipment are received.

It is obligatory on the part of the bidder to make available services of their testing and commissioning engineers (with necessary testing equipment) to assist the Board staff in commissioning their equipment as and when their services are requisitioned.

13.0 TROPICAL TREATMENT:

13.1 Under the climatic conditions prevalent at site, the equipment furnished under the specification will be subjected to operation under ambient temperatures of 50 deg. C and very high relative humidity. All equipment shall therefore, be suitably designed and treated for normal life and satisfactory operation under the hot and humid climatic conditions prevalent at site and shall be dust and vermin proof. All parts and surfaces which are subject to corrosion, shall be made of such material and shall be provided with such protective finishes as would protect the equipment installed from any injurious effects of excessive humidity. All electrical auxiliary equipment shall be specifically treated for tropical conditions, and the materials and methods for this treatment shall be approved in advance by the Purchaser.

14.0 FOUNDATIONS AND FIXTURES:

Foundations will be prepared by the Purchaser in accordance with the basic designed data to be supplied by the contractor. All fixtures necessary for proper erection of the embedded parts shall be supplied by the supplier.

15.0 SPARES:

The bidder shall quote for the spare parts as indicated in Annexure-II. The bidder shall quote for individual items as indicated in Annexure-II for the purpose of addition / deletion. The rates given in Annexure-II for individual items.

ANNEXURE

SCHEDULE OF DETAILED EQUIPMENT

LTAC PANELS**:**
THE PANEL SHALL BE EQUIPPED AND WIRED WITH THE FOLLOWING
1 No. Incoming circuit consisting of:
2 Nos. 3 Line Programmable Multi-function meters class 1.0 compatible with RS485 Modbus protocol to connect to SCADA/SAS system. Meters shall display all the parameters like currents, voltages, power and frequency etc.,
1 No. 500 Amps 30 kA moulded case circuit breaker TPN
3 Nos. 500 Amps HRC fuses
3 Nos. Air cooled ring type current transformer for metering purpose with a ratio 400-200/5 Amps of Class-I accuracy.
3 Nos. Pilot LED indicating lamps for incoming supply
3 Nos. Fuses for above lamps
1 Set 'TPN' electrolytic aluminum bus bars, 500 Amps rating of suitable size supported on non-hygroscopic SRBP supports.
2 Nos. Outgoing circuit each consisting of: 200A
1 No. 200 Amps 'TPN' switch
3 Nos. 200 Amps 'HRC' fuses.
2 Nos. Outgoing circuit each consisting of: 63A
1 No. 63 Amps 'TPN' switch
3 Nos. 63 Amps 'HRC' fuses.
6 Nos. Outgoing circuit each consisting of: 32A
1 No. 32 Amps 'TPN' switch
3 Nos. 32 Amps 'HRC' fuses.
16 Nos. Outgoing circuits each consisting of: 16A
1 No. 16 Amps single phase switch
1 No. 16 Amps HRC fuses
1 No. Over &under Voltage protection Relay with alaram and trip functions.
(However this will be finalized during detailed engineering while drawing approval.)
1 No. Neutral link
1 Set Cable glands and cable supports
1 Set Circuit designation label.
1 Set Earth Bus 6x 25mm copper
1 Set name plates for all the equipment mounted on panel (front side and inside)
1 Set Cubicle illumination consisting 10 watt LED lamp with door controlled switch.
1 Set Terminal block/containing terminals for outgoing circuits.
1 Set Cable lugs.
1 Set 15 Amps/5 Amps/2 Amps plug socket with ON/OFF switch.
Note: 2 sets of HRC Fuses of all ratings should be supplied extra as spare, along with each panel.
Note: The supplier shall include all such accessories which are required for proper functioning of the above LTAC panel. Such accessories even if they are not mentioned in the above list shall be deemed to have